

ECE 280 – Lab 1 Report

Signals and Systems: Arduino Signal Generator

Student Information

Course/Section: _____
Team Members: _____
Student IDs: _____
Lab Date: _____
TA/Instructor: _____

Submission Checklist

Submit this PDF report and supporting files (code + images) to your LMS.

- RC filter breadboard photo (labeled R, C, Arduino pin)
 - DMM photos for 0%, 50%, 100% duty cycles
 - Oscilloscope screenshot with time/div and volts/div visible
 - MATLAB plot screenshot (500 samples, labeled axes)
 - Arduino code and MATLAB script files
 - Jitter comparison histograms (Phase 2 vs Phase 3)
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1 Pre-Lab Results (Summary)

Q1: Filter design target ($f_c \approx 50.000$ Hz)

Chosen values: $R =$ _____ Ω , $C =$ _____ F

Computed cutoff: $f_c = \frac{1}{2\pi RC} =$ _____ Hz

Q2: RC response

$h(t)$ summary and settling-time derivation ($\tau = RC$):

Q3: Quantization and SQNR

$\Delta V =$ _____ V, SQNR (8-bit) = _____ dB

2 Phase 1: Hardware Assembly and Verification

1.1 RC Filter Assembly

PWM pin used: _____
_____ F

Measured R : _____ Ω

Measured C :

RC schematic used and wiring notes:

Breadboard photo:

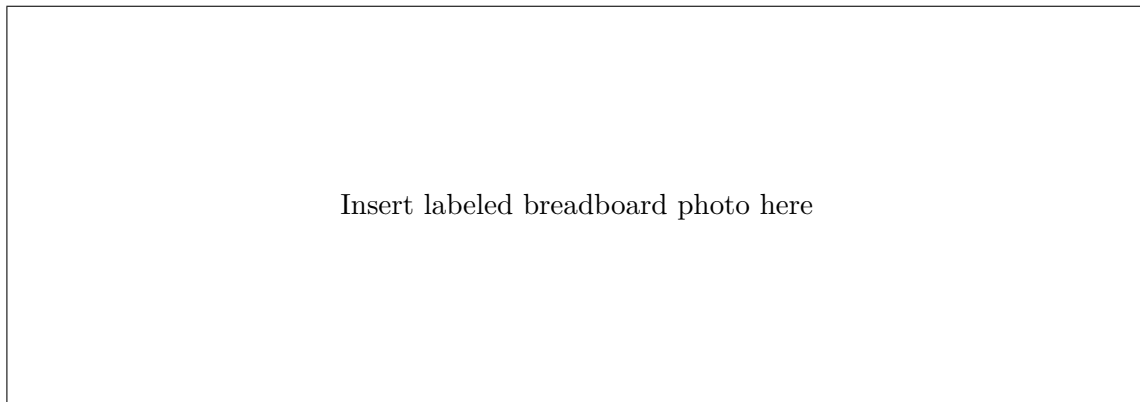


Figure 1: RC filter hardware implementation.

1.2 Baseband Verification (DMM)

Table 1: PWM Duty-Cycle Verification

Duty Cycle	Expected (V)	Measured (V)	Percent Error (%)
0%	0.00		
50%	2.50		
100%	5.00		

Linearity comment (is output proportional to duty cycle?):

3 Phase 2: Acquisition and MATLAB Integration

2.1 Waveform Generation

Assigned waveform: _____

Assigned frequency f_0 : _____ Hz

Implementation method (LUT/algorithmic/hybrid) and timing strategy:

2.2 Oscilloscope Capture

Table 2: Scope Measurements

Parameter	Value
Peak-to-Peak Voltage (V)	
Observed Frequency (Hz)	
Period (ms)	
Time/Div setting	
Volts/Div setting	
Visible distortion/ringing (Yes/No + note)	

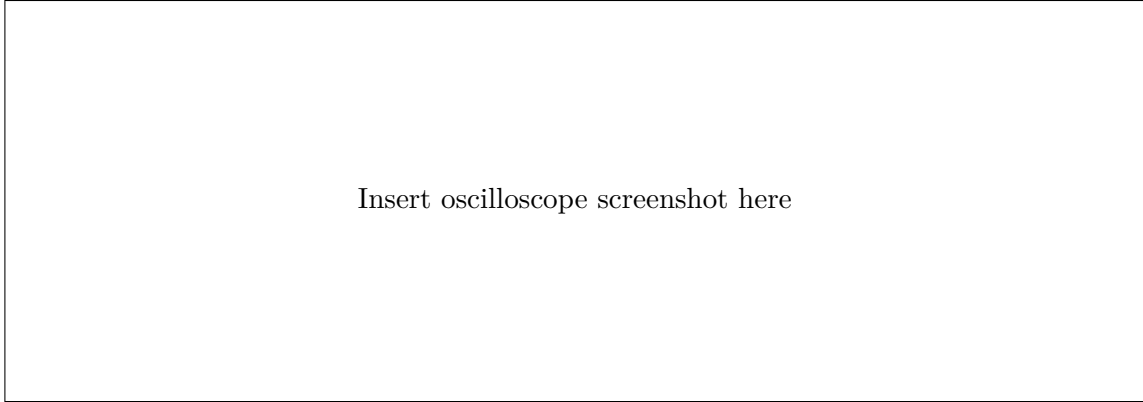


Figure 2: Oscilloscope capture of filtered waveform.

2.3 Serial Acquisition (MATLAB)

Serial port: _____

Baud: 115200

Samples acquired: 500

Measured sampling period T_s : _____ s
_____ Hz

Measured sampling frequency $f_s = 1/T_s$:

Any acquisition issues or fixes:

2.4 MATLAB Visualization

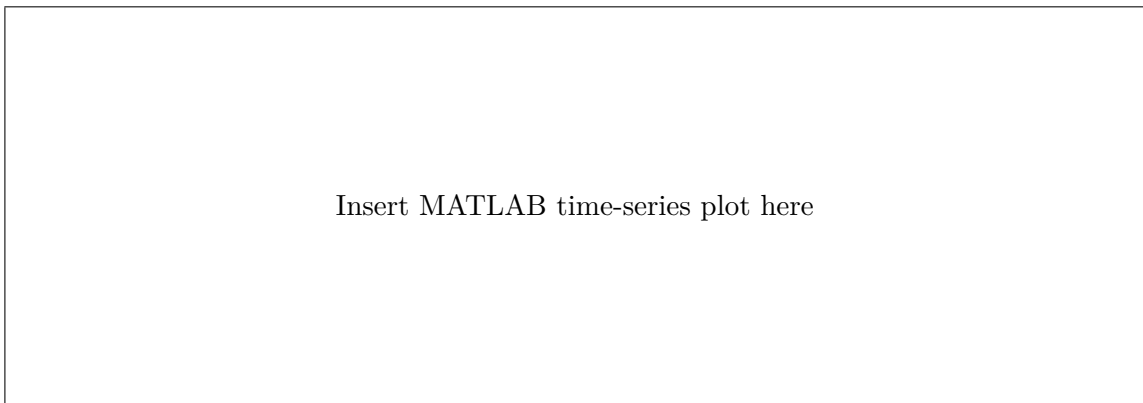


Figure 3: 500-sample MATLAB waveform plot with labeled axes and title.

4 Phase 3: Interrupt Optimization

3.1 Baseline Jitter (Delay-Based)

Table 3: Baseline Jitter Statistics (Phase 2)

Metric	Value (μs)
Mean interval	
Standard deviation	
Minimum interval	
Maximum interval	
Range	

3.2 Interrupt-Driven Implementation

Timer used: _____

ISR frequency target: _____ Hz

Brief ISR implementation summary and key register settings:

3.3 Jitter Comparison

Table 4: Jitter Improvement Summary

Metric	Phase 2 (Delay)	Phase 3 (Interrupt)
Mean interval (μs)		
Std. dev. (μs)		
Range (μs)		

Jitter improvement percentage:

$$\text{Improvement} = \frac{\sigma_{\text{phase2}} - \sigma_{\text{phase3}}}{\sigma_{\text{phase2}}} \times 100\% = \underline{\hspace{2cm}}$$

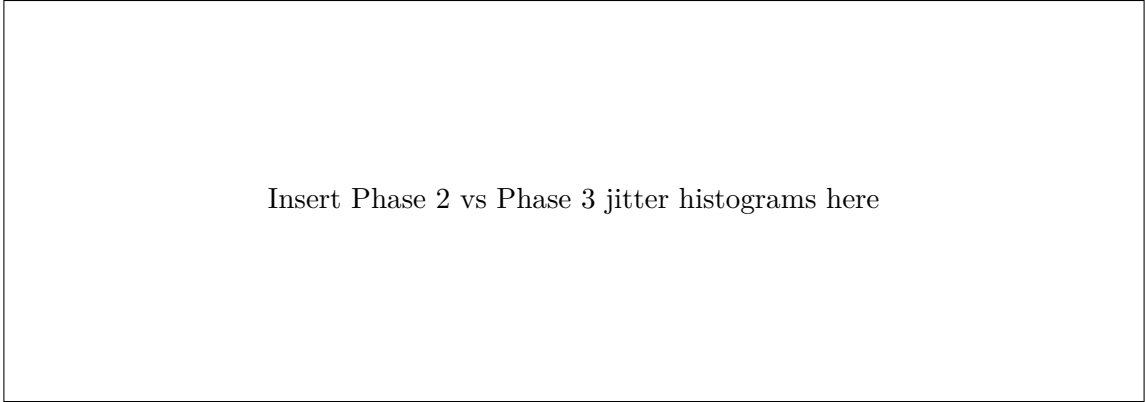


Figure 4: Jitter distribution comparison.

5 Cross-Validation and Error Analysis

Table 5: Theory vs Measurement Comparison

Parameter	Theoretical	Scope	MATLAB	Error (%)
Amplitude (V)				
Frequency (Hz)				
Period (ms)				
Peak-to-peak (V)				

Primary error sources (rank top 3):

- 1.
- 2.
- 3.

Discussion: Explain mismatch between theory and experiments, and comment on interrupt im-

pact.

6 Code Appendix

List submitted files and brief description.

File Name	Description

Academic Integrity Statement

We certify that this submission represents our own lab work and measurements, and that all submitted images/code correspond to our team's execution.

Signature(s): _____ Date: _____